

Lecture 1: Introduction
DD2421 Machine Learning

Part II: Classification problems and Nearest Neighbor rule

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In this part we will visit:

- Some more examples of applications
- Concept of classification
  - Hand-written digit recognition
- Simple approaches for classification
  - Nearest Neighbour method

Where is machine learning useful?

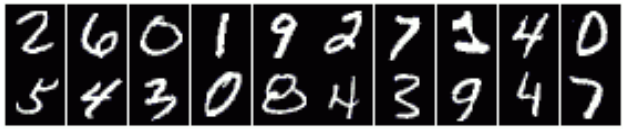
- A pattern exists.
- Data available for training.
- Hard/impossible to define rules mathematically.

Related terms on data analysis

- Pattern Recognition
- Data Mining
- Statistics

ML applications are pervasive

- Optic character recognition (OCR)



- Medical diagnosis
- DNA analysis



(by bejarprints)

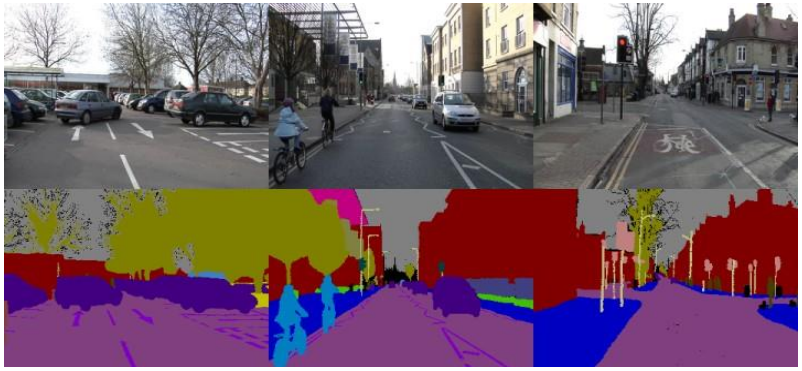
- Remote sensing
- Speech technology



- NLP / automatic translation

- Computer Vision
- Robotics
- Finance
- Recommender systems: books, movies
- Biometrics: fingerprint, iris, face
- Generative AI

…



(The Cambridge-driving Labeled Video Database)







Concept of classification

- We would like to enable a computer to learn from data to answer a question - “What is it?”

  You’re given sample data (for finding patterns).

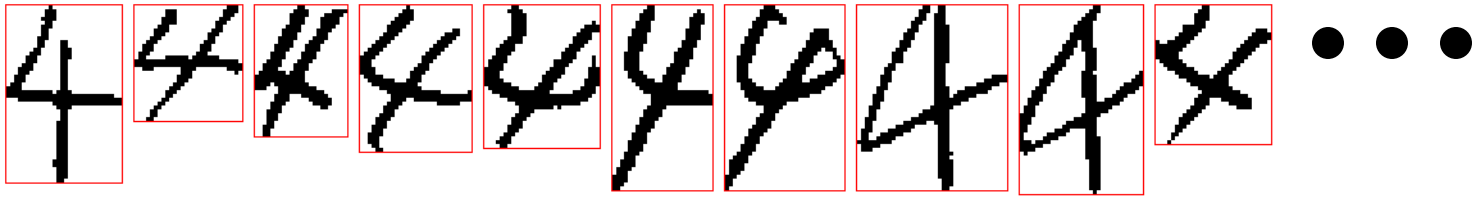
The framework of classification

1. Training phase: to give the concept of classes to a machine using labeled data
2. Testing phase: to determine the class of new unseen (unlabeled) data

Example: Hand-written digits

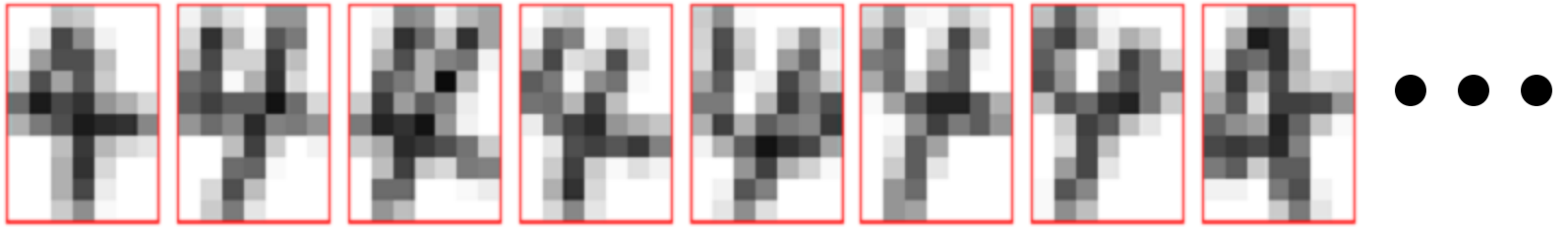
One of the first commercial system with ML, used for zip codes

Training samples:



Feature extraction

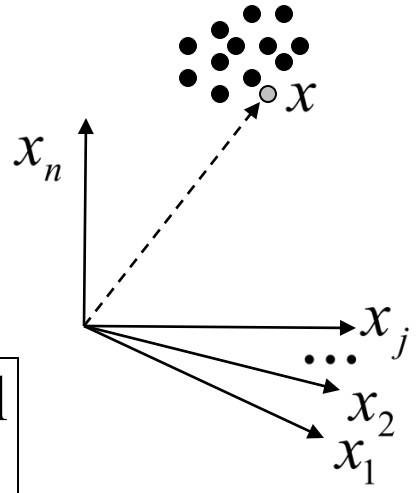
Pattern vectors: normalized (& blurred) patterns



Feature extraction for digit recognition

- Represent an image by a feature vector
  - Sampling: n = 7 x 10 pixels



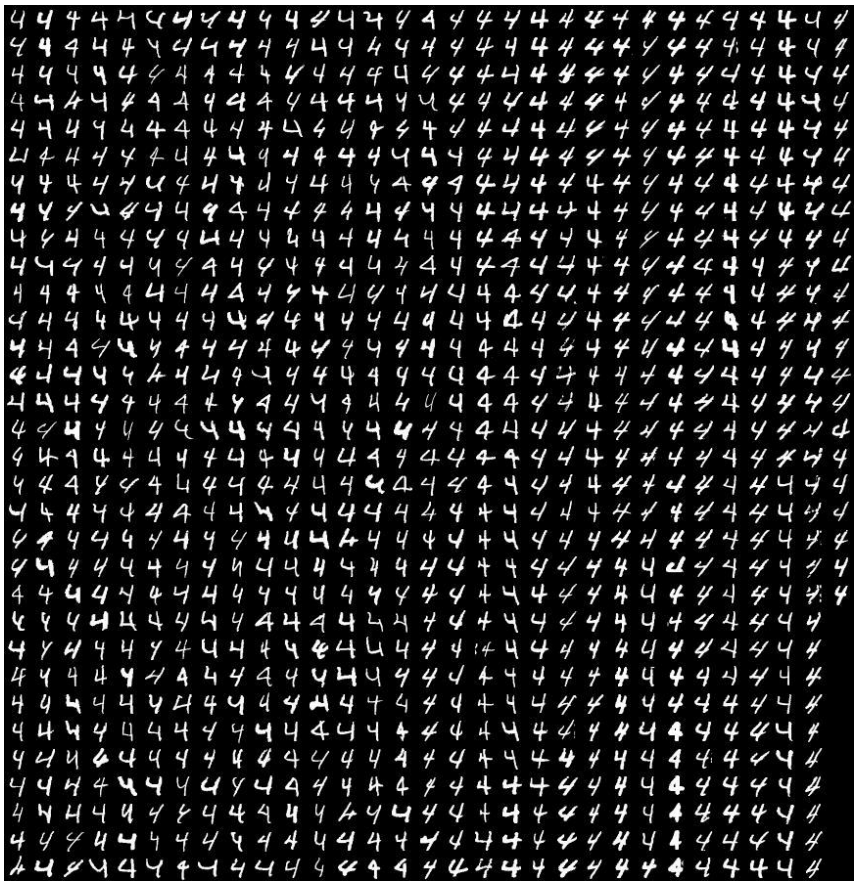


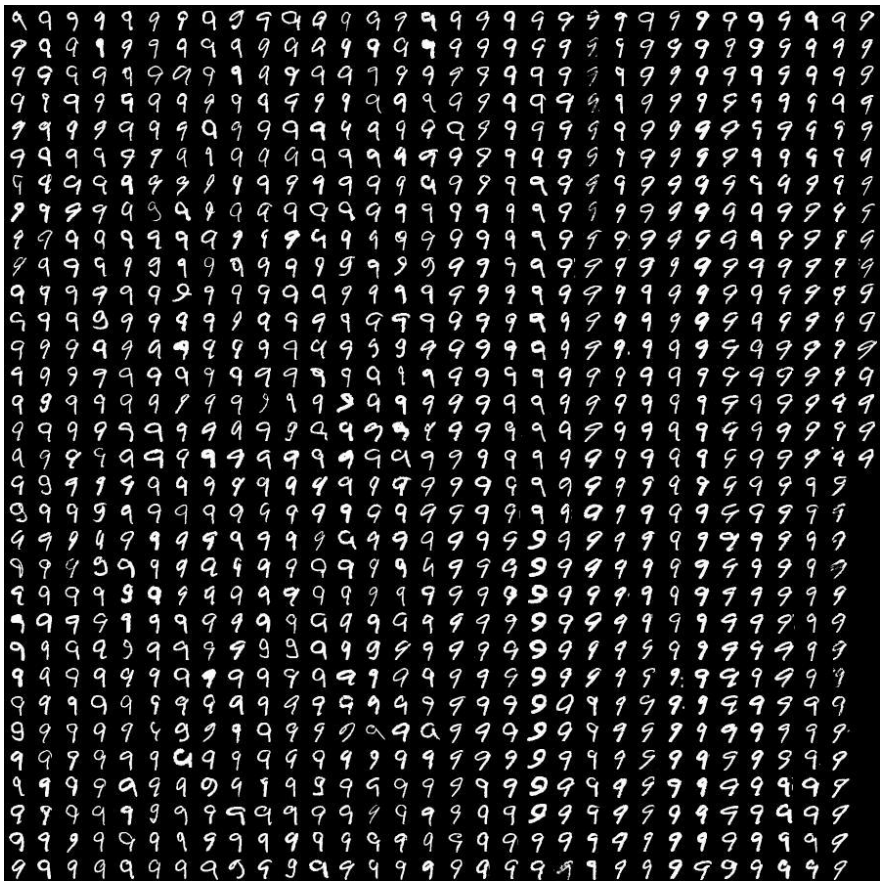
n-dimensional feature space

  - A set of n gray values: $x = (x_1, \ldots, x_n)$

    i.e. corresponding to a point in feature space

More training samples of “4” and others





http://yann.lecun.com/exdb/mnist/

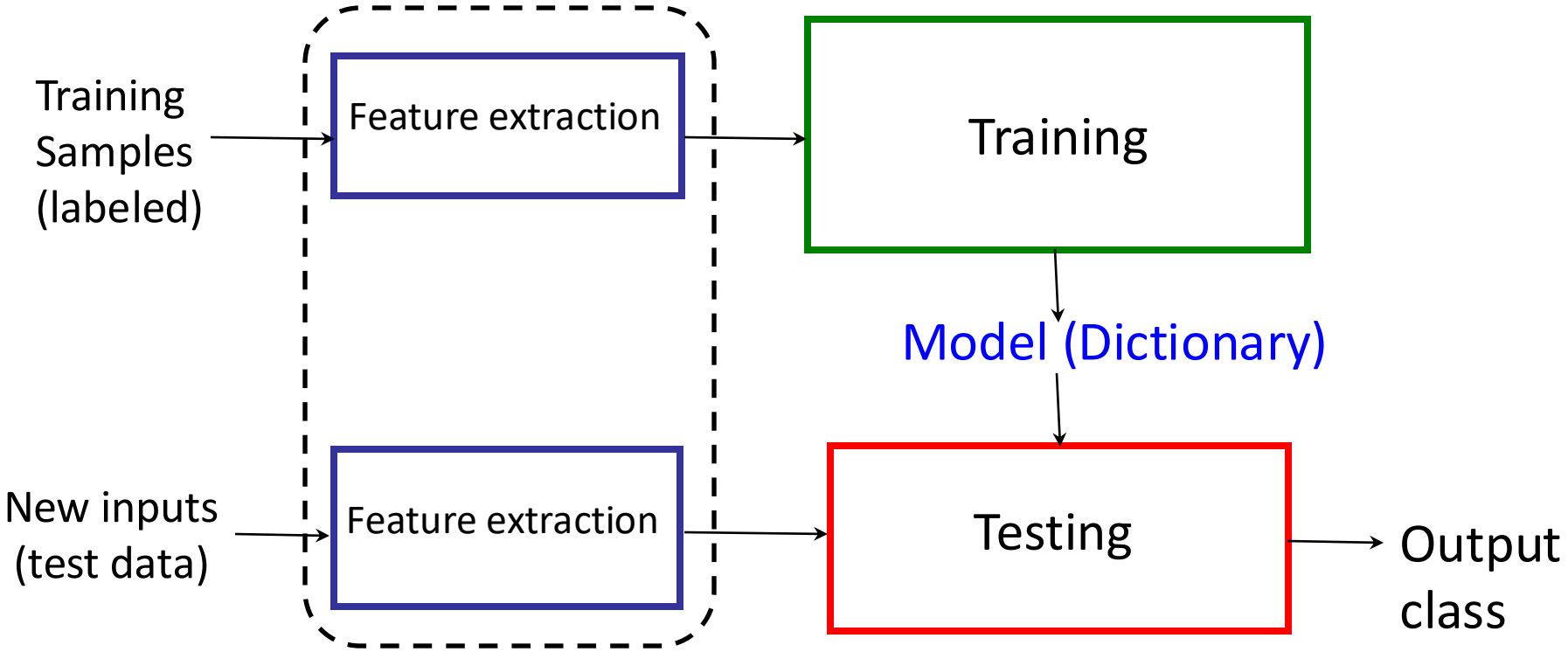
Example: Face images

Training samples of frontal faces



(Face image database, CMU)

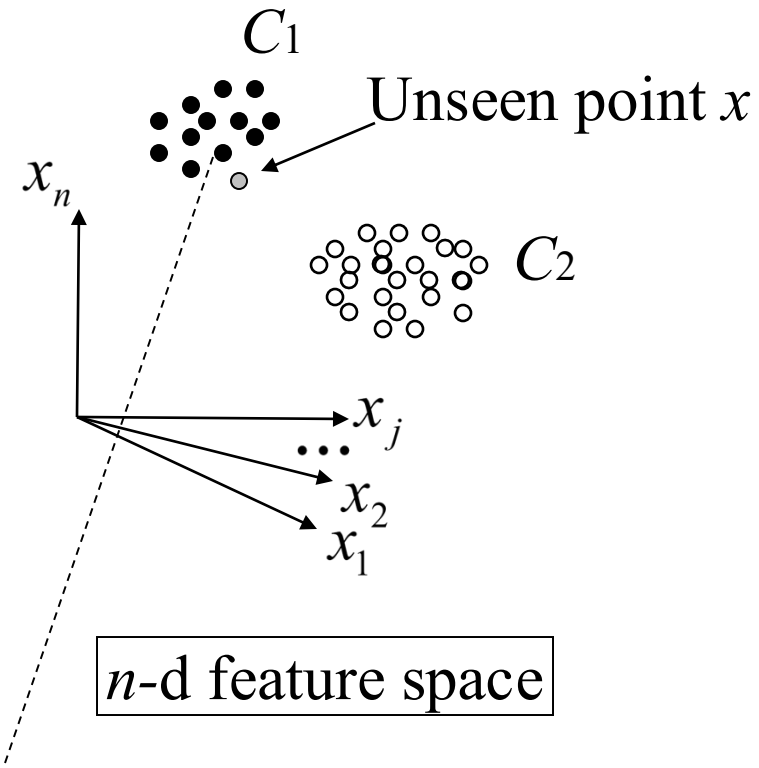
Schematic of classification



Nearest Neighbour methods

- Binary classification
  - N_1 samples of class C_1
  - N_2 samples of class C_2
  - Unseen data x

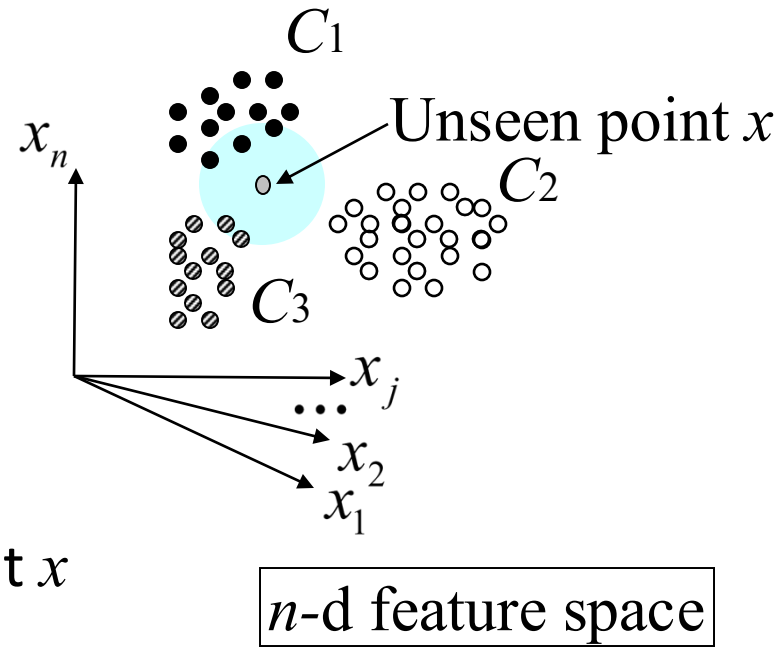
→ Compute distances to all the $N_1 + N_2$ samples



- Find the nearest neighbour

→ classify x to the same class

- k-nearest neighbour rule
  - Compute the distances to all the samples from new data x
  - Pick k neighbours that are nearest to x



→ Majority vote to classify point x

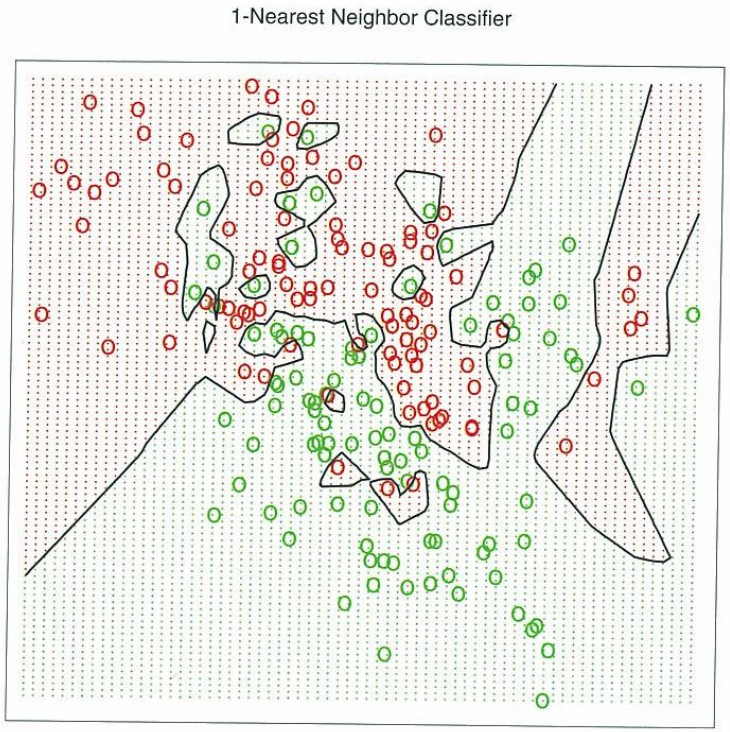
(Nearest Neighbour is 1-NN)

- How does k-NN compare to 1-NN ?

What is the influence of k ?

$k = 1$

$k = ?$



?

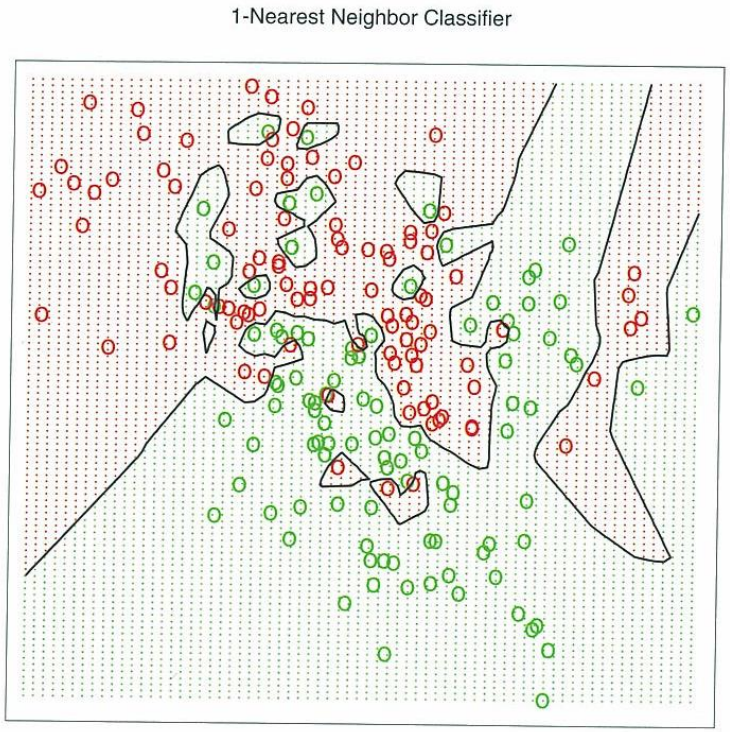
Decision boundaries with different k

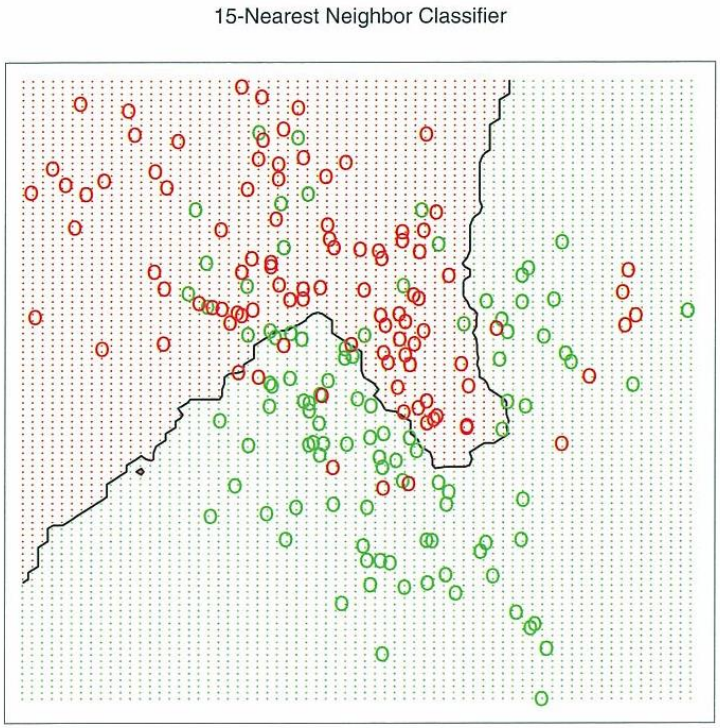
$k = 1$

No misclassifications on training data

$k = 15$

More generalized





(T. Hastie et al, The Elements of Statistical Learning)

Pros and cons of k-NN

- Pros:
  - simple; only with a single *hyper-parameter* k
  - applicable to multi-class problems
  - good performance, effective in low dimension data
- Cons:
  - costly to compute distances to search for the nearest
  - memory requirement: must store all the training set!

- k-NN / 1-NN comparison summary
  - the boundary becomes smoother as k increases
  - k-NN better generalizes given many samples

Notes on "**generalization**"

- Our goal is to determine the class of unseen data.
- Strategies/parameters that achieve minimum loss on training samples is not necessarily best for test data.
- We want the machine/model to learn the *true pattern* (and not noise) that resides in the sample data for generalization.

Keywords to remember

- Classification
  - Feature extraction
  - Training/Testing
  - Generalization

- Classification methods
  - Nearest Neighbour rule

- Decision trees (to come)